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Image replacement inpainting

Abstract

A method for replacing an object in an image. The method may include identifying a first object at a position within a first image, masking, based on the first image and the position of the first object, a target area to produce a masked image, generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a difference between the target area of training images and content of generated images at location corresponding to the target area of the training images, generating, based on the masked image and the second image, a third image, and adding, to the third image, a new object different from the first object.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Application under 35 U.S.C. § 371 and claims the benefit of International Application No. PCT/US2020/032677, filed May 13, 2020. The disclosure of the foregoing application is incorporated herein by reference in its entirety.

BACKGROUND

(2) This document relates to a method for replacing portions of an image using inpainting.

SUMMARY

(3) In general, one innovative aspect of the subject matter described in this specification may be embodied in a computer-implemented method for replacing an object in an image. The method may include identifying a first object at a position within a first image, masking, based on the first image and the position of the first object, a target area to produce a masked image, generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images, generating, based on the masked image and the second image, a third image, and adding, to the third image, a new object different from the first object.

(4) These and other embodiments may each optionally include one or more of the following features.

(5) In some implementations, the inpainting machine learning model is trained using a loss function that represents a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images.

(6) In some implementations, the first image is a frame of a video.

(7) In some implementations, generating, based on the masked image and the second image, a third image includes masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image, and generating, based on the masked image and the inverse masked image, the third image. The inverse target area can include an area of the second image that is outside of the location corresponding to the target area of the first image. Masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image can include generating, from the second image, an inverse masked image that includes at least some content of the second image that is inside the target area and that does not include at least some content of the second image that is outside the target area. Generating the third image can include compositing the inverse masked image with the masked image.

(8) In some implementations, the method includes extrapolating, based on the third image, a fourth image, wherein each of the first image, second image, third image, and fourth image is a frame of a video.

(9) Other embodiments of this aspect include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices.

(10) Particular embodiments of the subject matter described in this document can be implemented so as to realize one or more of the following advantages. In certain environments, there has previously been no way to replace portions of images in files such as final renderings of video or images without creating areas surrounding the portions to be replaced that are visually inconsistent with the original video or image. For example, closed captions in a final rendering of a video could not simply be altered or replaced to provide translations, corrections, or updates, as previous methods would generally replace the areas surrounding the original closed captions with a solid

shape of a single color or attempting to remove the original closed captions using inpainting techniques and rendering the translated captions over the box or the removed captions. These previous methods are jarring and/or inaccurate, and can negatively affect a user's experience when viewing the video or image and, by extension, the user's reaction to the content of the video or image. In general, existing solutions for inpainting are trained using entire images and may not effectively reconstruct smaller areas of an image due to variations in structure and detail across the image, and that shortcoming is addressed by the techniques, devices, and systems discussed herein. The techniques described in this document enable a system to provide an improved solution to image replacement by masking an area to be replaced, using an inpainting machine learning model (also referred to as an inpainting model) trained using areas surrounding particular types of objects within an image, to inpaint (e.g., predictively create or reconstruct) a proper subset of the image (e.g., corresponding to the masked area) that is smaller than the image, and rendering the replacement image over the inpainted area.

(11) This new system supports the replacement of portions of images in finally rendered videos or images having both natural and synthetic backgrounds. Because current inpainting model training techniques train a model using images that vary greatly in visual characteristics, existing inpainting methods produce inaccurate results when applied to images having areas that vary in structure, color, and texture, among other visual characteristics. This new image replacement method uses a deep learning model that is trained on reconstructing limited areas of images surrounding particular types of objects. For example, this new system can use an inpainting model trained to reconstruct areas surrounding simple, solid shapes when the target content is a simple solid shape, and an inpainting model trained to reconstruct areas surrounding text when the target content is text. Because the inpainting models are trained to focus on limited areas surrounding a target object (e.g., within a masked area), the models can more accurately reconstruct the details and variations within the area. More specifically, the loss function for the inpainting model can focus on the portion of the newly generated image corresponding to the masked area, rather than allowing the area outside the masked area to dominate the loss function, as the content outside the masked area should be readily reproducible leading to very low differences between those portions of the original image and the newly generated image. Furthermore, the inpainting models can be trained for particular types of objects such that inpainting models can be trained for reconstruction surrounding and within a complex object with multiple areas of transparency; text of a particular style and spacing; a particular image representing a logo; among other specific applications.

(12) The system provides an efficient solution that allows for videos and images to be altered and/or customized based on an original image after the image has been finally rendered. The system improves the accuracy with which a target area can be masked and removed from an image using an inpainting method by implementing a novel solution that trains inpainting models using tailored portions of test data to improve the accuracy with which the system reconstructs areas having various visual characteristics. Additionally, the system allows for dynamic updating of content based on particular characteristics, including user preferences. For example, if a user is watching a video that includes English language audio and closed captioning in Spanish, and the user's preferred language is Chinese, the system allows the video to be altered and re-rendered such that the video includes closed captioning in Chinese. In some implementations, the system can automatically detect the alterations or customizations needed. For example, the system can determine, based on user information indicating that the user's preferred language is Chinese, to replace the closed captioning in English with generated closed captioning in Chinese. In some implementations, the system is provided with the alterations or customizations to be made. For example, the system can receive a set of closed captioning in Chinese and an indication that the object to be replaced is the set of closed captioning in Spanish. Furthermore, non-text content can also be replaced, as desired. For example, the video can be modified to replace items that are deemed inappropriate for the current audience, or that violate terms of service for distributing

videos (e.g., excessively graphic content). In some implementations, the system allows content to be moved relative to its frame. For example, a logo in a top layer that is obscuring content in the background can be relocated so that the content in the background is visible.

(13) Leveraging this method allows images and videos to be customized for a particular application, user preferences, and geographical locations, among other factors. In addition, the improved removal and re-rendering process provides efficiencies in video and image content production by reducing the overall amount of resources consumed by allowing a single video project to be adapted and used for different applications and users. This allows reduced bandwidth of transmission of video across networks as only one video project need be transferred to a node, and reduced processing resource required to render more than one video project as the aspects of a video project being modified by the system are significantly less than a whole new video project being rendered (and then transferred to a node). This system provides an avenue for video designers, developers, and content providers to tailor image and video content after the design, editing, and post-processing phases, which require heavy expenditure of resources.

(14) Additionally, the benefits of this method include allowing video creators to alter only a portion of a video project. The method allows video projects to be customized for a particular application, user preferences, and geographical locations, among other factors. In addition to reducing the amount of resources required to customize and render a video for each request, the system reduces the overall amount of resources consumed by allowing a single video project to be adapted and used for different applications and users.

(15) The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an example environment for improved image replacement inpainting.

(2) FIG. 2 depicts a data flow of an improved image replacement inpainting process.

(3) FIG. 3 is a diagram of an example machine learning process for training an image replacement inpainting model.

(4) FIG. 4 is a flow chart of an example process for improved image replacement inpainting.

(5) FIG. 5 is a block diagram of an example computing system.

(6) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(7) This document describes methods, systems, and devices that improve content replacement in images and videos, enabling a single image or video to be used for multiple applications, modified to comply with local norms and standards, comply with terms of service for image/video distribution, and to be tailored to different users' preferences.

(8) In some implementations, the system allows a finally rendered image or video to be altered at the time of request and customized based on the context in which the image or video will be presented, e.g., geographic area to which the image will be distributed and/or audience information. In some situations, the techniques discussed herein can be used by a video creator who wishes to adapt a video project for a particular client.

(9) For example, the system can receive a finally rendered video from a content provider. The video includes an image that is to be adapted based on a viewer's location. In this example, the video is a public service announcement to be used across the United States, and the image can be a logo

displayed in a corner of each frame of the video. The logo can be adapted to be the state in which a viewer is located. The system can detect the logo as a target, the target's position within the frame, that the target is a solid shape, and determine a bounding box that surrounds the target. The bounding box can be determined based on parameters including a target accuracy granularity and a target image quality, among other parameters. For each frame, the system can then generate a mask for the portion of the frame within the bounding box that removes the logo and an “inverse” mask for the portion of the frame outside of the bounding box. The mask for the portion of the frame within the bounding box is applied to the frame to create a masked image with the content inside of the bounding box removed. The masked image is provided to an inpainting model trained to focus on areas surrounding the bounding box. The inpainting model is used to generate a second, different frame that includes the content inside of the bounding box as well as the content outside of the bounding box. Thus, the inpainting model used to generate the content is adapted for this particular application. The inverse mask for the portion of the frame outside of the bounding box is applied to the second, different frame to create an inverse masked image with the content outside of the bounding box removed. The system can then render a new frame by compositing the masked image and the inverse masked image. In other words, the portion of the original image that is outside of the bounding box can be combined with the portion of the generated image that is inside the bounding box to create a new image that does not include the target. A replacement object is identified and can be placed in the new frame in the same position in which the target was located in the original frame. In some implementations, the replacement object is placed in the new frame in a different position relative to the position in which the target was located in the original frame. The system can render the new frame with the replacement object and render the entire video with new frames to produce a final video to be presented in response to the request.

(10) FIG. 1 is a block diagram of an example environment **100** for improved image replacement inpainting. The example environment **100** includes a network **102**, such as a local area network (LAN), a wide area network (WAN), the Internet, or a combination thereof. The network **102** connects electronic document servers **104** (“Electronic Doc Servers”), user devices **106**, and a digital component distribution system **110** (also referred to as DCDS **110**). The example environment **100** may include many different electronic document servers **104** and user devices **106**.

(11) A user device **106** is an electronic device that is capable of requesting and receiving resources (e.g., electronic documents) over the network **102**. Example user devices **106** include personal computers, mobile communication devices, and other devices that can send and receive data over the network **102**. A user device **106** typically includes a user application, such as a web browser, to facilitate the sending and receiving of data over the network **102**, but native applications executed by the user device **106** can also facilitate the sending and receiving of data over the network **102**.

(12) One or more third parties **150** include content providers, product designers, product manufacturers, and other parties involved in the design, development, marketing, or distribution of videos, products, and/or services.

(13) An electronic document is data that presents a set of content at a user device **106**. Examples of electronic documents include webpages, word processing documents, portable document format (PDF) documents, images, videos, search results pages, and feed sources. Native applications (e.g., “apps”), such as applications installed on mobile, tablet, or desktop computing devices are also examples of electronic documents. Electronic documents **105** (“Electronic Docs”) can be provided to user devices **106** by electronic document servers **104**. For example, the electronic document servers **104** can include servers that host publisher websites. In this example, the user device **106** can initiate a request for a given publisher webpage, and the electronic document server **104** that hosts the given publisher webpage can respond to the request by sending machine Hyper-Text Markup Language (HTML) code that initiates presentation of the given webpage at the user device **106**.

(14) Electronic documents can include a variety of content. For example, an electronic document **105** can include static content (e.g., text or other specified content) that is within the electronic document itself and/or does not change over time. Electronic documents can also include dynamic content that may change over time or on a per-request basis. For example, a publisher of a given electronic document can maintain a data source that is used to populate portions of the electronic document. In this example, the given electronic document can include a tag or script that causes the user device **106** to request content from the data source when the given electronic document is processed (e.g., rendered or executed) by a user device **106**. The user device **106** integrates the content obtained from the data source into a presentation of the given electronic document to create a composite electronic document including the content obtained from the data source.

(15) In some situations, a given electronic document can include a digital content tag or digital content script that references the DCDS **110**. In these situations, the digital content tag or digital content script is executed by the user device **106** when the given electronic document is processed by the user device **106**. Execution of the digital content tag or digital content script configures the user device **106** to generate a request **108** for digital content, which is transmitted over the network **102** to the DCDS **110**. For example, the digital content tag or digital content script can enable the user device **106** to generate packetized data request including a header and payload data. The request **108** can include data such as a name (or network location) of a server from which the digital content is being requested, a name (or network location) of the requesting device (e.g., the user device **106**), and/or information that the DCDS **110** can use to select digital content provided in response to the request. The request **108** is transmitted, by the user device **106**, over the network **102** (e.g., a telecommunications network) to a server of the DCDS **110**.

(16) The request **108** can include data specifying the electronic document and characteristics of locations at which digital content can be presented. For example, data specifying a reference (e.g., URL) to an electronic document (e.g., webpage) in which the digital content will be presented, available locations (e.g., digital content slots) of the electronic documents that are available to present digital content, sizes of the available locations, positions of the available locations within a presentation of the electronic document, and/or media types that are eligible for presentation in the locations can be provided to the DCDS **110**. Similarly, data specifying keywords designated for the selection of the electronic document ("document keywords") or entities (e.g., people, places, or things) that are referenced by the electronic document can also be included in the request **108** (e.g., as payload data) and provided to the DCDS **110** to facilitate identification of digital content items that are eligible for presentation with the electronic document.

(17) Requests **108** can also include data related to other information, such as information that the user has provided, geographic information indicating a state or region from which the request was submitted, or other information that provides context for the environment in which the digital content will be displayed (e.g., a type of device at which the digital content will be displayed, such as a mobile device or tablet device). User-provided information can include demographic data for a user of the user device **106**. For example, demographic information can include age, gender, geographical location, education level, marital status, household income, occupation, hobbies, social media data, and whether the user owns a particular item, among other characteristics.

(18) For situations in which the systems discussed here collect personal information about users, or may make use of personal information, the users may be provided with an opportunity to control whether programs or features collect personal information (e.g., information about a user's social network, social actions or activities, profession, a user's preferences, or a user's current location), or to control whether and/or how to receive content from the content server that may be more relevant to the user. In addition, certain data may be anonymized in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be anonymized so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a

city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over how information is collected about him or her and used by a content server.

(19) Data specifying characteristics of the user device **106** can also be provided in the request **108**, such as information that identifies a model of the user device **106**, a configuration of the user device **106**, or a size (e.g., physical size or resolution) of an electronic display (e.g., touchscreen or desktop monitor) on which the electronic document is presented. Requests **108** can be transmitted, for example, over a packetized network, and the requests **108** themselves can be formatted as packetized data having a header and payload data. The header can specify a destination of the packet and the payload data can include any of the information discussed above.

(20) The DCDS **110** selects digital content that will be presented with the given electronic document in response to receiving the request **108** and/or using information included in the request **108**. In some implementations, the DCDS **110** is implemented in a distributed computing system (or environment) that includes, for example, a server and a set of multiple computing devices that are interconnected and identify and distribute digital content in response to requests **108**. The set of multiple computing devices operate together to identify a set of digital content that is eligible to be presented in the electronic document from among a corpus of millions or more of available digital content. The millions or more of available digital content can be indexed, for example, in a digital component database **112**. Each digital content index entry can reference the corresponding digital content and/or include distribution parameters (e.g., selection criteria) that condition the distribution of the corresponding digital content.

(21) In some implementations, digital components from digital component database **112** can include content provided by third parties **150**. For example, digital component database **112** can receive, from a third party **150** that uses machine learning and/or artificial intelligence to navigate public streets, a photo of a public intersection. In another example, digital component database **112** can receive, from a third party **150** that provides services to bicycle riders, specific questions to which third party **150** would like responses from users. Additionally, the DCDS **110** can present video content, including content from, for example, video rendering processor **130** and video content provided by video rendering processor **130** stored in the digital component database **112**.

(22) The identification of the eligible digital content can be segmented into multiple tasks that are then assigned among computing devices within the set of multiple computing devices. For example, different computing devices can each analyze a different portion of the digital component database **112** to identify various digital content having distribution parameters that match information included in the request **108**.

(23) The DCDS **110** aggregates the results received from the set of multiple computing devices and uses information associated with the aggregated results to select one or more instances of digital content that will be provided in response to the request **108**. In turn, the DCDS **110** can generate and transmit, over the network **102**, reply data **114** (e.g., digital data representing a reply) that enables the user device **106** to integrate the select set of digital content into the given electronic document, such that the selected set of digital content and the content of the electronic document are presented together at a display of the user device **106**.

(24) Image replacement processor **120** receives, or retrieves, images and videos and replaces particular portions of an image or video. Image replacement processor **120** includes an object processor **122**, an inpainting model **124**, and a rendering processor **126**. Object processor **122** processes an image or video to detect objects and generate bounding boxes surrounding the detected objects. Inpainting model **124** uses machine learning techniques to recreate particular portions of an image or video within, for example, the bounding boxes generated by object processor **122**. Rendering processor **126** renders altered images and or videos.

(25) For ease of explanation, object processor **122**, inpainting model **124**, and rendering processor **126** are shown in FIG. 1 as separate components of image replacement processor **120**. Image

replacement processor **120** can be implemented as a single system on non-transitory computer-readable media. In some implementations, one or more of object processor **122**, inpainting model **124**, and rendering processor **126** can be implemented as integrated components of a single system. Image replacement processor **120**, its components object processor **122**, inpainting model **124**, and rendering processor, and their respective operations and outputs are described in further detail below.

(26) Inpainting model **124** can be a statistical and/or machine learning model. For example, inpainting model **124** can be a generative-adversarial-network (GAN) model that is trained using a loss function that focuses on a limited area of an image (e.g., within a masked area or within some specified distance of the masked area). In some implementations, inpainting model **124** can include, in addition to the GAN model, any of a variety of models such as decision trees, linear regression models, logistic regression models, neural networks, classifiers, support vector machines, inductive logic programming, ensembles of models (e.g., using techniques such as bagging, boosting, random forests, etc.), genetic algorithms, Bayesian networks, etc., and can be trained using a variety of approaches, such as deep learning, association rules, inductive logic, clustering, maximum entropy classification, learning classification, etc. In some examples, the machine learning models may use supervised learning. In some examples, the machine learning models use unsupervised learning. The training of inpainting model **124** is described in detail below with respect to FIG. 3.

(27) The techniques described below enable a system to efficiently edit and render video, regardless of complexity.

(28) FIG. 2 shows an example data flow **200** of an improved image replacement inpainting process in the example environment of FIG. 1. Operations of data flow **200** are performed by various components of the system **100**. For example, operations of data flow **200** can be performed by image replacement processor **120** in communication with DCDS **110** and/or user device **106**.

(29) The flow begins with step A, in which image replacement processor **120** receives a final rendering **202**. Final rendering **202** can be an image or a video. For example, image replacement processor **120** can receive a final image rendering **202** from a content provider, such as a third party **130**, or from DCDS **110**. In some implementations, image replacement processor **120** receives final rendering **202** from a user such as a user of a user device **106**. In some implementations, image replacement processor **120** retrieves final rendering **202** from a storage medium such as a database. For example, image replacement processor **120** retrieves final rendering **202** from digital component database **112**. In some implementations, image replacement processor **120** can receive instructions to retrieve final rendering **202**.

(30) The flow continues with step B, in which image replacement processor **120** detects a target object to be replaced and generates a bounding box **206** that surrounds the target object. For example, object processor **122** can detect target object **204** and its object type. In this particular example, target object **204** is a text label. In other implementations, the target object can be other types of content such as text, an open shape, a closed shape, a complex shape with transparency parameters, a simple shape, a moving shape, and/or a changing shape, among other types of objects.

(31) Object processor **122** can also generate a bounding box **206** around the target object **204**. In some implementations, the bounding box **206** is determined based on predetermined parameters. For example, the bounding box **206** can be determined to be a rectangular shape surrounding the target object having margins of at least a predetermined size between the bounding box **206** and the target object **204**. For example, object processor **122** can determine a bounding box **206** around the target object **204** by detecting the edges of target object **204** and following a margin parameter. The margin parameter can be provided, for example, by a user of a user device **106**, by a content provider, by the DCDS **110**, and/or by a third party **130**.

(32) Object processor **122** also detects target object **204**'s position within final rendering **202**. For

example, object processor **122** can determine target object **204**'s position in terms of an x position and a y position. In some implementations, object processor **122** can determine target object **204**'s position relative to a reference point (e.g., a reference pixel location and an area occupied by the target object). In some implementations, object processor **122** can determine target object **204**'s absolute position.

(33) The flow continues with step C, in which image replacement processor **120** recreates the content data of the area of final rendering **202** within the bounding box **206**. First, in substep C-1, image replacement processor **120** generates a mask for the area of final rendering **202** within the bounding box **206** to remove the content of final rendering **202** within the bounding box **206**. Image replacement processor **120** then applies the mask to final rendering **202** to generate a masked image **208** that does not include the content of final rendering **202** within the bounding box **206**. In some implementations, object processor **122** generates and applies the mask to final rendering **202** to generate a masked image **208**.

(34) Next, in substep C-2, inpainting model **124** can recreate the content within the area surrounding the target object (e.g., the area within the bounding box **206**) by uniformly sampling $r \in [r_{\text{sub.min}}, r_{\text{sub.max}}]$ where $r_{\text{sub.min}}$ and $r_{\text{sub.max}}$ are the minimum and maximum aspect ratios, respectively. Inpainting model **124** can then uniformly sample $h \in$

$$(35) [\sqrt{\frac{a_{\text{min}}}{r}}, \min(\sqrt{\frac{a_{\text{max}}}{r}}, y_{\text{max}} - 2m, \frac{x_{\text{max}} - 2m}{r})]$$

where $a_{\text{sub.min}}$ and $a_{\text{sub.max}}$ are the minimum and maximum coverage areas, respectively; $x_{\text{sub.max}}$ and $y_{\text{sub.max}}$ are the image width and height, respectively; and m is the desired margin around the edges of the target object. Inpainting model **124** can next set $w=rh$ to conform with a selected aspect ratio. In some implementations, the selected aspect ratio is the aspect ratio of the original image. In other implementations, the aspect ratio can be selected based on user input or other parameters. Inpainting model **124** can then uniformly sample $x_{\text{sub.corner}} \in [m, x_{\text{sub.max}} - (m+w)]$ and $y_{\text{sub.corner}} \in [m, y_{\text{sub.max}} - (m+h)]$.

(36) Using this sampled data, inpainting model **124** can recreate content within the area of final rendering **202** inside of the bounding box **206** without the target object **204**. Inpainting model **124** can perform, for example, structural or geometric inpainting, textural inpainting, and/or a combination of structural and textural inpainting. Because inpainting model **124** is able to adapt to images with structural properties ranging from complex, variable textures and boundaries to simple edges and structural properties. For example, inpainting model **124** can generate a second image **210** using the masked image that includes the content within the area of final rendering **202** recreated by inpainting model **124**. In other words, the second image can be a recreation of the final rendering **202** generated by the inpainting model **124** using the masked image **208** as input.

(37) In some implementations, second image **210** includes the content of final rendering **202** outside of the bounding box **206** exactly as represented in final rendering **202**. For example, the content of final rendering **202** outside of the bounding box **206** can be used in second image **210** exactly as it is input, or can be reproduced by inpainting model **124** exactly as it was input to inpainting model **124**. In some implementations, second image **210** includes content that represents the content of final rendering **202** outside of the bounding box **206** that is generated by inpainting model **124**. For example, inpainting model **124** can generate the content of final rendering **202** outside of the bounding box **206**, and this generated content may not represent the content of final rendering **202** exactly as it was input. The second image **210** can have the same dimensions as the final rendering **202**. In some implementations, the second image **210** can have different dimensions from the final rendering **202**. For example, the second image **210** can be a scaled version of the final rendering **202**.

(38) The flow continues with step D, in which image replacement processor **120** generates an “inverse” masked image. First, in substep D-1, image replacement processor **120** generates an “inverse” mask for the area of the second image **210** outside of the position of a bounding box **212**.

Image replacement processor **210** can generate the bounding box **212** at the same position and with the same proportions as the bounding box **206** in the final rendering **202**. The bounding box **212** can contain the recreated content generated by inpainting processor **124**. Inpainting model **124** can, for example, generate an inverse mask for the area of the second image **210** outside of the bounding box **212** to remove the area of the second image **210** outside of the bounding box **212**. Inpainting model **124** then applies the inverse mask to the second image **210** to create an inverse masked image **214**. In some implementations, object processor **122** generates and applies the mask to second image **210** to generate an inverse masked image **214**.

(39) In substep D-2, inpainting model **124** can then create a composite of the masked image **208** and the inverse masked image **210** to generate a third image that represents a version of final rendering **202** that does not include the target object **204**. For example, inpainting model **124** can use alpha blending to combine the masked image **208** and the inverse masked image **210** into a third image **216** that does not include the target object **204**. Other techniques for combining the masked image **208** and inversed masked image **210** can also be used to create the third image **216**, which includes a portion of each of the masked image **208** and the inversed masked image **210**.

(40) The flow continues with step E, in which the DCDS **110** receives a request **108** for content from user device **106**. Request **108** is transmitted by user device **106** to DCDS **110** when the client device interacts with digital content. For example, if a user of user device **106** clicks a link to download a shopping application, the link can cause user device **106** to transmit a request **108** to DCDS **110**. The request **108** can include interaction tracking data from the client device **106**. For example, the request **108** can include tracking data such as an indication of the interaction, the digital content with which user device **106** interacted, and an identifier that uniquely identifies user device **106**. In some implementations, the request **108** includes an indication of the provider of the digital content and the location of the destination server that hosts the requested resource.

(41) The DCDS **110** processes the request and forwards a request for a particular image or video, including customization information, to image replacement processor **120**. For example, the DCDS **110** can select a particular video using the content selection process described above with respect to FIG. **1** and determine customization information to provide to image replacement processor **120**. In some implementations, the DCDS **110** determines customization information based on of the context in which the image or video will be presented at a user device **106**. In some implementations, the DCDS **110** determines customization information based on information from, for example, a third party **130** that provides content to the DCDS **110**.

(42) The flow continues with step F, in which image replacement processor **120** adds a replacement object to the third image **216**, or the composite image. The replacement object **218** can be of the same type as the target object **204** or of a different type. For example, the replacement object **218** can be an image while the target object **204** is text. In some implementations, the replacement object **218** can be selected based on the customization information. In some implementations, the replacement object **218** is provided to the image replacement processor **120**. Object processor **122** can add a replacement object **218** in the same position in which the target object **204** was located in final rendering **202**. In some implementations, object processor **122** can add a replacement object **218** in a different position relative to the position in which the target object **204** was located in final rendering **202**. In one particular example, object processor **122** can move the target object **204** to a different position in the third image **216** relative to the position in which the target object **204** was located in the final rendering **202**.

(43) The flow continues with step G, in which image replacement processor **120** re-renders the image with the added object to generate a new image. For example rendering processor **126** can create a composite of the third image **216** and the replacement object **218** into a new rendering **220**. In some implementations, rendering processor **126** can use alpha blending techniques to create new rendering **220**.

(44) In some implementations, image replacement processor **120** can create new rendering **220** by

creating a composite of the second image **210** and the replacement object **218**. For example, image replacement processor **120** may not perform step D.

(45) The flow continues with step H, in which image replacement processor **120** provides the new image or video to DCDS **110** for presentation to user device **106** along with, or as, reply **114**. For example, image replacement processor **120** provides new rendering **220** to DCDS **110** for presentation to user device **106** along with reply **114**. As described above, reply data **114** can indicate the image or video content as re-rendered by image replacement processor **120** in addition the requested electronic document. Reply data **114** is transmitted by DCDS **110** to user device **106** in response to DCDS **110** receiving request **108** and determining, based on the received distribution parameters and user data indicated in request **108**, that the distribution parameters are satisfied.

(46) FIG. 3 is a diagram of an example block diagram of a system **300** for training an intelligent garment system. For example, system **300** can be used to train inpainting model **124** as described with respect to FIG. 1.

(47) Examples **202** are provided to training module **210** as input to train, in this particular example, a GAN-based inpainting model. Examples **202** can be positive examples (i.e., examples of inpainted area masks that have a threshold accuracy level) or negative examples (i.e., examples of inpainted areas that have less than a threshold accuracy level). Examples **202** can include the ground truth areas, or an area determined to have a threshold accuracy level with respect to an actual, correct area from an original image.

(48) The ground truth indicates the actual, correct area from an original image. For example, examples **202** can include training sets in which an actual area is removed from an original image and provided to training module **210**. In some implementations, a ground truth area can be generated and provided to training module **210** as an example **202** by generating, using inpainting model **124**, an inpainted area mask and confirming that inpainted area mask has a threshold level of accuracy with respect to the actual, correct area from the original image. In some implementations, a human can manually verify the accuracy of the inpainted area mask. The inpainted area mask can be automatically detected and labelled by pulling data from a data storage medium that contains verified inpainted area masks.

(49) The ground truth future motion can be correlated with particular inputs of examples **202** such that the inputs are labelled with the ground truth. With ground truth labels, training module **210** can use examples **202** and the labels to verify model outputs of inpainting model **124** and continue to train the model to improve the accuracy with which the model generates inpainted area masks.

(50) Training module **210** trains inpainting model **124** using a targeted area loss function **212**. Training module **110** uses targeted area loss function **212** to train inpainting model **124** to generate an inpainted area mask. Targeted area loss function **212** can account for variables such as a predicted texture and/or a predicted structure, among other variables. For example, targeted area loss function **212** can focus on an area surrounding a target object, such the area within bounding box **206** surrounding target object **204** as depicted with respect to FIG. 2.

(51) For example, inpainting model **124** can use a loss function, such as for example the one specified as follows:

$$(52) G_{\text{optimal}} = \underset{G}{\operatorname{argmin}} \underset{D}{\operatorname{max}} L(G, D)$$

$$L(G, D) = E_{x,y} [\log D(x, y)] + E_x [\log(1 - D(x, G(x)))] + E_{x,y} [y - G(x)]$$

This loss function represents a difference between a target area of the first image and content of the second image at a location corresponding to the target area of the first image. Because the content outside the masked area is highly reproducible, and in some implementations can be copied exactly, leading to very low differences between those portions of the original image and the newly generated image, the system's loss function focuses on the portion of the newly generated image corresponding to the masked area instead of allowing the area outside the masked area to dominate the loss function.

(53) Training module **210** can train inpainting model **124** manually or the process could be automated. Training module **210** uses the loss function **212** and examples **202** labelled with the ground truth areas to train inpainting model **124** to learn what is important for the model. For example, training module **210** can train inpainting model **124** by optimizing the model based on the loss function. In some implementations, optimizing the model based on the loss function includes minimizing the result of the loss function, which minimizes a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images. In some implementations, training module **210** can use a threshold value for the difference, such as a maximum percentage difference. For example, training module **210** can train inpainting model **124** to reconstruct content at a location corresponding to a target area in a training image with a maximum of 5%, 10%, or 20% difference, among other percentages. In some implementations, training module **210** can train inpainting model **124** to reconstruct content at a location corresponding to a target area in a training image with a threshold number of pixels, such as a maximum number of different pixels or a minimum number of similar, or identical pixels. Training module **210** allows inpainting model **124** to learn by changing the weights applied to different variables to emphasize or deemphasize the importance of the variable within the model. By changing the weights applied to variables within the model, training module **210** allows the model to learn which types of information (e.g., which textures, structural components, boundaries, etc.) should be more heavily weighted to produce a more accurate inpainting model.

(54) The examples and variables can be weighted based on, for example, feedback from image replacement processor **120**. For example, if image replacement processor **120** collects feedback data that indicates that the loss function **212** is producing results that do not have a threshold level of accuracy with respect to a particular type of target object, then inpainting model **124** can weigh particular configurations more heavily than other configurations. The inpainted areas generated by inpainting model **124** that satisfy a threshold level of accuracy can be stored as positive examples **202**.

(55) Inpainting model **124** can use various types of models, including general models that can be used for all types of target objects and properties of original image, or final rendering **202**, and customized models that can be used for particular subsets of target objects or original images sharing a set of characteristics, and can dynamically adjust the models based on the detected type of target object or original image. For example, the classifier can use a base GAN for all target objects and then tailor the model to each object.

(56) Inpainting model **124** can adapt to and generate masks for a vast number of types of target objects. In some implementations, the training process for inpainting model **124** is a closed loop system that receives feedback information and dynamically updates the inpainting model network based on this feedback information.

(57) FIG. 4 is a flow chart of an example process **400** for process for improved image replacement inpainting. In some implementations, process **400** can be performed by one or more systems. For example, process **400** can be implemented by image replacement processor **120**, DCDS **110**, user device **106**, and third party **140** of FIGS. 1-3. In some implementations, the process **400** can be implemented as instructions stored on a non-transitory computer readable medium, and when the instructions are executed by one or more servers, the instructions can cause the one or more servers to perform operations of the process **400**.

(58) The process **400** for replacing an object in an image begins with identifying a first object at a position within a first image (**402**). For example, image replacement processor **120** receives a final rendering **202** and identifies a target object **204** at a particular position within the first image. Object processor **122** of image replacement processor **120** can perform the identification. Object processor **122** can also identify an object type of target object **204** and determine a position of target object **204** in final rendering **202**. Object processor **122** can generate a bounding box **206** surrounding target object **204**.

(59) The process **400** continues with masking, based on the first image and the position of the first object, a target area to produce a masked image (**404**). For example, image replacement processor **120** can generate a mask to remove the area within bounding box **206** based on final rendering **202**. Object processor **122** or inpainting model **124** can perform the mask generation and application process as described above with respect to FIGS. 1-3.

(60) The process **400** continues with generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a loss function that represents a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images (**406**). For example, inpainting model **124** can generate a second image **210** based on final rendering **202** as described above with respect to FIGS. 1-3. As described above, inpainting model **124** is trained to focus on areas that are a subset of an entire image and surrounding target objects of particular types.

(61) The process **400** continues with generating, based on the masked image and the second image, a third image (**408**). For example, image processor **120** can generate an inverse masked image based on the second image and an inverse mask as described above with respect to FIGS. 1-3. Image processor **120** can then generate a third image by compositing the masked area and the inverse mask. For example, inpainting model **124** or object processor **122** can generate third image **216**.

(62) The process **400** continues with adding, to the third image, a new object (**410**). For example, image processor **120** can generate a new rendering **220** by adding a replacement object **218** that is different from target object **206** to third image **216**. Rendering processor **126** can use, for example, alpha blending techniques to create new rendering **222**. In some implementations, image processor **120** adds replacement object **218** to third image **216** at the relative position of the target object **206** within the final rendering **202**. In some implementations, image processor **120** adds replacement object **218** to third image **216** at a different position. In some implementations, replacement object **218** is the same object as the target object **206** and the process **400** relocates the target object **206** with respect to its position within final rendering **202**.

(63) FIG. 5 is block diagram of an example computer system **500** that can be used to perform operations described above. The system **500** includes a processor **510**, a memory **520**, a storage device **530**, and an input/output device **540**. Each of the components **510**, **520**, **530**, and **540** can be interconnected, for example, using a system bus **550**. The processor **510** is capable of processing instructions for execution within the system **500**. In one implementation, the processor **510** is a single-threaded processor. In another implementation, the processor **510** is a multi-threaded processor. The processor **510** is capable of processing instructions stored in the memory **520** or on the storage device **530**.

(64) The memory **520** stores information within the system **500**. In one implementation, the memory **520** is a computer-readable medium. In one implementation, the memory **520** is a volatile memory unit. In another implementation, the memory **520** is a non-volatile memory unit.

(65) The storage device **530** is capable of providing mass storage for the system **500**. In one implementation, the storage device **530** is a computer-readable medium. In various different implementations, the storage device **530** can include, for example, a hard disk device, an optical disk device, a storage device that is shared over a network by multiple computing devices (e.g., a cloud storage device), or some other large capacity storage device.

(66) The input/output device **540** provides input/output operations for the system **500**. In one implementation, the input/output device **540** can include one or more network interface devices, e.g., an Ethernet card, a serial communication device, e.g., and RS-232 port, and/or a wireless interface device, e.g., and 802.11 card. In another implementation, the input/output device can include driver devices configured to receive input data and send output data to other input/output devices, e.g., keyboard, printer and display devices **560**. Other implementations, however, can also

be used, such as mobile computing devices, mobile communication devices, set-top box television client devices, etc.

(67) Although an example processing system has been described in FIG. 5, implementations of the subject matter and the functional operations described in this specification can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(68) An electronic document (which for brevity will simply be referred to as a document) does not necessarily correspond to a file. A document may be stored in a portion of a file that holds other documents, in a single file dedicated to the document in question, or in multiple coordinated files.

(69) Embodiments of the subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described in this specification can be implemented as one or more computer programs, i.e., one or more modules of computer program instructions, encoded on computer storage media (or medium) for execution by, or to control the operation of, data processing apparatus. Alternatively, or in addition, the program instructions can be encoded on an artificially generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate physical components or media (e.g., multiple CDs, disks, or other storage devices).

(70) The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

(71) The term “data processing apparatus” encompasses all kinds of apparatus, devices, and machines for processing data, including, by way of example, a programmable processor, a computer, a system on a chip, or multiple ones, or combinations, of the foregoing. The apparatus can include special-purpose logic circuitry, e.g., an FPGA (field-programmable gate array) or an ASIC (application-specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(72) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a standalone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, subprograms, or portions of code). A computer program can be deployed to be executed on one computer or on

multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(73) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special-purpose logic circuitry, e.g., an FPGA (field-programmable gate array) or an ASIC (application-specific integrated circuit).

(74) Processors suitable for the execution of a computer program include, by way of example, both general and special-purpose microprocessors. Generally, a processor will receive instructions and data from a read-only memory or a random-access memory or both. The essential elements of a computer are a processor for performing actions in accordance with instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS) receiver, or a portable storage device (e.g., a universal serial bus (USB) flash drive), to name just a few. Devices suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including, by way of example, semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto-optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special-purpose logic circuitry.

(75) To provide for interaction with a user, embodiments of the subject matter described in this specification can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(76) Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back-end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front-end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).

(77) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. In some embodiments, a server transmits data (e.g., an HTML page) to a client device (e.g., for purposes of displaying data to and receiving user input from a user interacting with the client device). Data generated at the client device (e.g., a result of

the user interaction) can be received from the client device at the server.

(78) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular embodiments of particular inventions. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(79) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(80) Thus, particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

Claims

1. A computer-implemented method for replacing an object in an image comprising: identifying a first object at a position within a first image; masking, based on the first image and the position of the first object, a target area to produce a masked image; generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images; generating, based on the masked image and the second image, a third image, wherein generating the third image comprises: masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image; and generating, based on the masked image and the inverse masked image, the third image; and adding, to the third image, a new object different from the first object.
2. The computer-implemented method of claim 1, wherein the first image is a frame of a video.
3. The computer-implemented method of claim 1, wherein the inpainting machine learning model is trained using a loss function that represents a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images.
4. The computer-implemented method of claim 1, wherein the inverse target area comprises an area of the second image that is outside of the location corresponding to the target area of the first image.
5. The computer-implemented method of claim 1, wherein masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image comprises generating, from the second image, an inverse masked image that

includes at least some content of the second image that is inside the target area and that does not include at least some content of the second image that is outside the target area.

6. The computer-implemented method of claim 1, wherein generating the third image comprises compositing the inverse masked image with the masked image.

7. The computer-implemented method of claim 1, further comprising extrapolating, based on the third image, a fourth image, wherein each of the first image, second image, third image, and fourth image is a frame of a video.

8. A system comprising: one or more processors; and one or more memory elements including instructions that, when executed, cause the one or more processors to perform operations including: identifying a first object at a position within a first image; masking, based on the first image and the position of the first object, a target area to produce a masked image; generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images; generating, based on the masked image and the second image, a third image, wherein generating the third image comprises: masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image; and generating, based on the masked image and the inverse masked image, the third image; and adding, to the third image, a new object different from the first object.

9. The system of claim 8, wherein the inpainting machine learning model is trained using a loss function that represents a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images.

10. The system of claim 8, wherein the first image is a frame of a video.

11. The system of claim 8, wherein the inverse target area comprises an area of the second image that is outside of the location corresponding to the target area of the first image.

12. The system of claim 8, wherein masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image comprises generating, from the second image, an inverse masked image that includes at least some content of the second image that is inside the target area and that does not include at least some content of the second image that is outside the target area.

13. The system of claim 8, wherein generating the third image comprises compositing the inverse masked image with the masked image.

14. The system of claim 8, the operations further comprising extrapolating, based on the third image, a fourth image, wherein each of the first image, second image, third image, and fourth image is a frame of a video.

15. A non-transitory computer storage medium encoded with instructions that when executed by a distributed computing system cause the distributed computing system to perform operations comprising: identifying a first object at a position within a first image; masking, based on the first image and the position of the first object, a target area to produce a masked image; generating, based on the masked image and an inpainting machine learning model, a second image different from the first image, the inpainting machine learning model being trained using a difference between the target area of training images and content of generated images at a location corresponding to the target area of the training images; generating, based on the masked image and the second image, a third image, wherein generating the third image comprises: masking, based on the second image and the location corresponding to the target area of the first image, an inverse target area to produce an inverse masked image; and generating, based on the masked image and the inverse masked image, the third image; and adding, to the third image, a new object different from the first object.

16. The non-transitory computer storage medium of claim 15, wherein the inpainting machine learning model is trained using a loss function that represents a difference between the target area

of training images and content of generated images at a location corresponding to the target area of the training images.

17. The non-transitory computer storage medium of claim 15, wherein the first image is a frame of a video.
